

ENTERPRISE INFRASTRUCTURE PLAN

for

ABC Startup Solutions

ITEP 414 – System Administration and Maintenance
Week 2 Portfolio Project – Individual Portfolio Project
Bachelor of Science in Information Technology

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Part 1 — Company Profile

Company Name: ABC Startup Solutions

Nature of Business: A software development startup offering custom web and mobile application development, SaaS products, and IT consulting services for small and medium-sized enterprises (SMEs).

Company Vision: To become a trusted software innovation partner across Southeast Asia, delivering technology that helps businesses grow securely and efficiently.

Company Mission: To design, build, and maintain reliable software solutions through a culture of continuous learning, technical excellence, collaboration, and integrity — while providing employees the infrastructure they need to do their best work.

Office Location (Fictional): 4th Floor, Tech Hub Tower, Filinvest Corporate City, Alabang, Muntinlupa City, Metro Manila, Philippines

Organizational Structure

ABC Startup Solutions operates on a flat, function-based structure appropriate for a 20-person startup. The Chief Executive Officer (CEO) oversees four department heads, each responsible for a functional unit:

- Chief Executive Officer (CEO) — overall company direction and strategy
- IT Manager — leads the Information Technology department (systems, infrastructure, and development support)
- HR Manager — leads the Human Resources department (recruitment, employee relations, payroll coordination)
- Finance Manager — leads the Finance department (budgeting, accounting, invoicing)
- Sales Manager — leads the Sales department (client acquisition, account management)

Each manager reports directly to the CEO, and each department's staff report to their respective manager. This structure keeps communication short and decision-making fast — critical for a startup preparing its first IT infrastructure rollout.

Employee Distribution

Department	Employees	% of Workforce
Information Technology	5	25%
Human Resources	4	20%
Finance	5	25%
Sales	6	30%
TOTAL	20	100%

Part 2 — Enterprise Hardware Inventory

The following hardware inventory equips all 20 employees with appropriate computing devices, plus the shared infrastructure needed to support daily operations. Quantities include a small buffer for spares/onboarding where relevant.

Asset ID	Hardware	Qty	Department	Purpose
HW-001	Desktop Computer (Intel i5, 16GB RAM, 512GB SSD)	3	Information Technology	Development and system administration workstations
HW-002	Desktop Computer (Intel i5, 8GB RAM, 256GB SSD)	4	Human Resources	Employee records, payroll, and office productivity tasks
HW-003	Desktop Computer (Intel i5, 8GB RAM, 256GB SSD)	5	Finance	Accounting, invoicing, and financial reporting
HW-004	Laptop (Intel i5, 16GB RAM, 512GB SSD)	2	Information Technology	Onsite/offsite troubleshooting and system deployment
HW-005	Laptop (Intel i5, 8GB RAM, 256GB SSD)	6	Sales	Client meetings, presentations, and field mobility
HW-006	Server (Rack-mount, Xeon, 64GB RAM, RAID storage)	1	Information Technology	Hosts file sharing, internal applications, and backups
HW-007	Router (Enterprise-grade Gigabit)	1	Information Technology	Internet gateway and traffic routing
HW-008	Switch (24-port managed)	2	Information Technology	Core and access-layer LAN connectivity
HW-009	Printer (Network Multifunction)	3	HR / Finance / Sales	Shared printing, scanning, and photocopying
HW-010	UPS (1500VA)	3	Information Technology	Power protection for server, router, and switch
HW-011	Wireless Access Point (WiFi 6)	2	Information Technology	Wireless coverage for laptops and mobile devices
HW-012	NAS Storage (4-bay, 8TB usable)	1	Information Technology	Centralized file storage and scheduled backups
HW-013	External Backup Drive (4TB USB 3.0)	2	Information Technology	Offline/offsite backup rotation
HW-014	Monitor (24-inch Full HD)	24	All Departments	Dual monitors for IT/Finance; single monitor for HR/Sales

Justification Highlights

- IT staff receive higher-spec desktops/laptops (i5, 16GB RAM) since they run development tools, virtual machines, and diagnostic software.

- Sales staff are issued laptops instead of desktops for mobility during client visits and off-site presentations.
- Three shared network printers are distributed by usage volume (HR, Finance, Sales) rather than one per department, reducing cost while meeting demand.
- Dual monitors are provided to IT and Finance staff, who benefit most from extended screen real estate (code/terminal windows, spreadsheets).
- UPS units protect the three most critical points of failure: the server, router, and switch, preventing data loss during power interruptions.

Part 3 — Enterprise Software Inventory

Software licensing follows a 20-seat baseline for productivity tools, with specialized development tools reserved for the IT department's workstations.

Software	Version	License	Purpose
Windows 11 Pro	23H2	Volume License (20 seats)	Primary operating system for all desktops and laptops
Ubuntu Server	22.04 LTS	Free / Open Source	Server OS for hosting internal applications and services
Microsoft Office	365 Business Standard	Subscription (20 seats)	Word processing, spreadsheets, email, and collaboration
VS Code	1.9x (latest)	Free / Open Source	Primary code editor for the development team
Git	2.4x	Free / Open Source	Version control for source code management
GitHub Desktop	3.x	Free	Graphical interface for Git/GitHub workflows
VirtualBox	7.x	Free / Open Source (GPL)	Virtual machines for testing and sandboxing
Google Chrome	Latest	Free	Standard web browser for daily use
Microsoft Defender	Built-in	Included with Windows 11 Pro	Endpoint antivirus and threat protection
AnyDesk	Latest	Business License (5 seats)	Remote support and troubleshooting by IT staff
7-Zip	23.x	Free / Open Source	File compression and archive management

Why Each Software Is Needed

- Windows 11 Pro: required (not Home) for domain join, BitLocker, and Remote Desktop support — all needed for centralized IT management.
- Ubuntu Server: a stable, free, and widely supported OS to host the company's internal file server and future web applications.
- Microsoft Office 365: provides company email (Outlook), document collaboration, and cloud storage in one subscription.
- VS Code, Git, and GitHub Desktop: the core toolchain for the development team to write, version, and publish code.
- VirtualBox: allows developers and IT staff to safely test configurations without affecting production machines.
- Google Chrome: the standard browser for web-based tools (CRM, cloud consoles, project management).
- Microsoft Defender: built-in, license-free endpoint protection — a practical first layer of antivirus for a startup budget.

- AnyDesk: enables the IT department to remotely support employees without needing to be physically present.
- 7-Zip: free and reliable compression tool for handling large file transfers and backups.

Part 4 — Enterprise Network Inventory

This inventory lists the physical networking components required to connect all four departments to the internet and to each other, following a standard small-office star topology.

Component	Specification	Qty	Purpose
ISP Modem	Fiber ONT, up to 1 Gbps	1	Terminates the internet connection from the ISP
Router	Enterprise-grade Gigabit router	1	Routes traffic between the LAN and the internet
Firewall	UTM Firewall (IDS/IPS enabled)	1	Perimeter security and traffic filtering
Switch	24-port managed Gigabit switch	2	Core and departmental network connectivity
Access Point	Dual-band WiFi 6 access point	2	Wireless coverage across the office floor
Patch Panel	24-port CAT6 patch panel	1	Organized termination point for structured cabling
CAT6 Cables	305m box, solid copper	3	Structured cabling to all workstations
RJ45 Connectors	Box of 100, shielded	2	Termination of CAT6 cable runs

Part 5 — Enterprise Network Diagram

The diagram below shows the logical network topology for ABC Startup Solutions: internet traffic enters through the ISP modem, passes through the router and firewall for security enforcement, then reaches the core switch, which distributes connectivity to the server, wireless access point, network printers, and — via the patch panel — to all four department LANs.

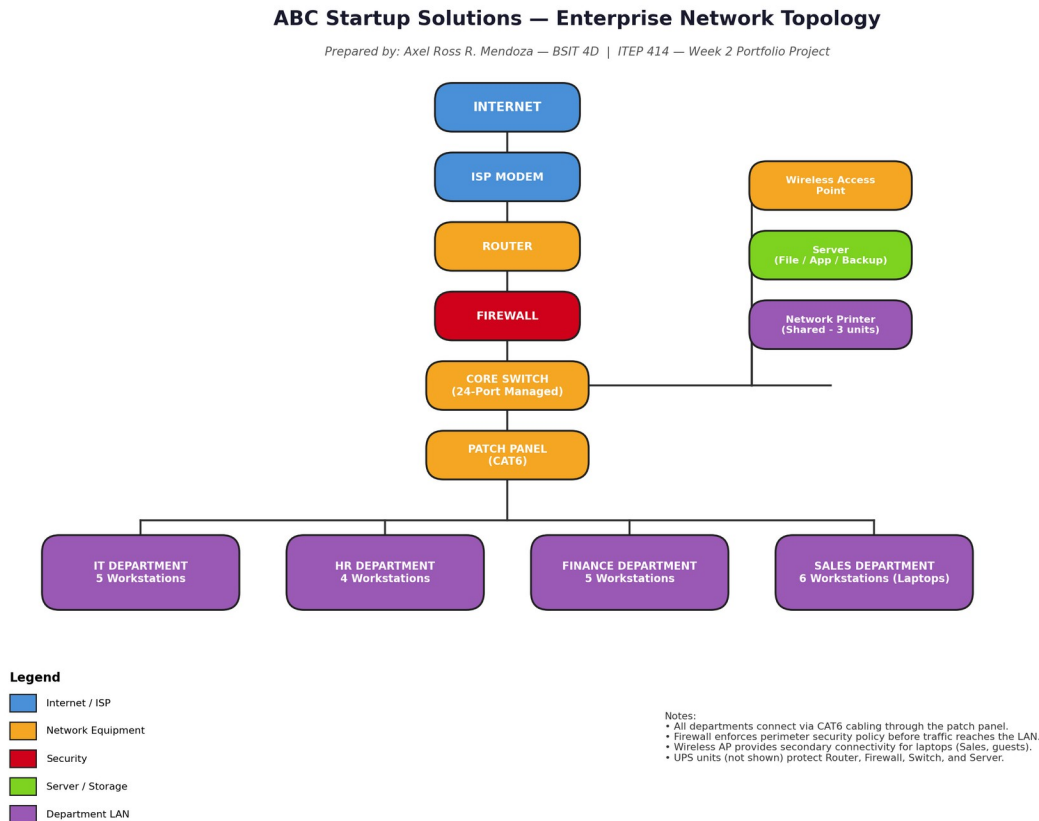


Figure 1. ABC Startup Solutions network topology, prepared by Axel Ross R. Mendoza (BSIT 4D). Exported as PNG and PDF; source files are provided separately in /diagrams.

Part 6 — System Administration Roles

Helpdesk Technician

Responsibilities: Provides first-line technical support to end users, resolves hardware/software issues, sets up new workstations, manages support tickets, and escalates complex problems to senior IT staff.

Skills: Troubleshooting, customer service, basic networking, OS installation, ticketing systems, patience under pressure.

Common Tools: Remote support software (AnyDesk, TeamViewer), ticketing systems (Zendesk, Freshservice), Active Directory basics, Microsoft Office.

Certifications: CompTIA A+, CompTIA IT Fundamentals (ITF+), Microsoft 365 Certified: Fundamentals.

Network Administrator

Responsibilities: Designs, configures, and maintains the company's network infrastructure — routers, switches, firewalls, and wireless access points — while monitoring performance and security.

Skills: Routing and switching, VLAN configuration, firewall management, network monitoring, subnetting, VPN setup.

Common Tools: Cisco IOS / MikroTik RouterOS, Wireshark, PRTG or Zabbix (monitoring), pfSense/FortiGate firewalls.

Certifications: CompTIA Network+, Cisco CCNA, Fortinet NSE.

Linux System Administrator

Responsibilities: Installs, configures, and maintains Linux-based servers; manages user accounts, permissions, cron jobs, backups, and system security hardening.

Skills: Shell scripting (Bash), package management, systemd, log analysis, storage/RAID management, service configuration (SSH, Apache/Nginx).

Common Tools: Bash/Terminal, SSH, cron, Ansible, Docker, Nagios/Zabbix for monitoring.

Certifications: CompTIA Linux+, Red Hat Certified System Administrator (RHCSA), LPIC-1.

Cloud Administrator

Responsibilities: Manages the organization's cloud infrastructure — provisioning virtual machines, storage, and networking resources, controlling access, and optimizing cloud costs.

Skills: Cloud architecture, Identity and Access Management (IAM), infrastructure-as-code, cost optimization, cloud security best practices.

Common Tools: AWS Management Console / Azure Portal / Google Cloud Console, Terraform, CloudWatch/Azure Monitor.

Certifications: AWS Certified Solutions Architect – Associate, Microsoft Azure Administrator (AZ-104), Google Associate Cloud Engineer.

How These Four Professionals Work Together

In a growing organization like ABC Startup Solutions, these four roles form a layered support system rather than working in isolation. The Helpdesk Technician is the first point of contact, resolving everyday user issues

and gathering the details needed to escalate anything beyond basic troubleshooting. When a problem involves connectivity or infrastructure, it moves to the Network Administrator, who keeps the routers, switches, and firewall running and secure. Anything involving the internal server — user accounts, file shares, or automated backups — falls to the Linux System Administrator, who keeps that machine stable and patched. As the company grows and begins relying on cloud-hosted services for scalability, the Cloud Administrator extends this same reliability to virtual infrastructure, managing access and costs so the on-premises and cloud environments work as one coherent system. Together, they form a chain of escalation and specialization that lets a lean 20-person startup run enterprise-grade IT without needing a large in-house team.

Part 7 — Infrastructure Recommendations

Internet Provider

A business-grade fiber connection (e.g., PLDT Enterprise, Globe Business, or Converge Business) with a minimum of 100 Mbps symmetrical bandwidth and a Service Level Agreement (SLA) guaranteeing uptime and priority support. A business plan is recommended over a residential plan because it includes static IP addresses (needed for remote access and future server hosting) and faster repair response times — both of which a growing software company cannot operate without.

Server Specifications

A rack-mounted server with an Intel Xeon or AMD EPYC processor, at least 64GB of ECC RAM, and RAID 1 or RAID 5 storage for redundancy is recommended. ECC RAM and RAID protect against data corruption and drive failure — both realistic risks once the server holds shared files, source code repositories, and backups for the whole company. Ubuntu Server LTS is recommended as the OS for its long-term stability and low licensing cost, which matters for a startup's budget.

Backup Strategy

A 3-2-1 backup strategy is recommended: 3 copies of data, on 2 different storage types, with 1 copy stored offsite (or in the cloud). In practice this means daily automated backups from the server to the NAS, weekly backups copied to external drives for offsite storage, and monthly verification restores to confirm backups are actually recoverable. This protects the company against hardware failure, ransomware, and accidental deletion — all common causes of data loss in small businesses.

Security Recommendations

Recommended measures include a properly configured UTM firewall with intrusion detection/prevention enabled, network segmentation (separate VLANs for staff devices, servers, and guest Wi-Fi), disabling unused ports and services, and enforcing automatic OS/software patching. These reduce the company's attack surface significantly and are inexpensive relative to the cost of a breach.

Antivirus

Microsoft Defender for Business is recommended as the baseline endpoint protection, since it is already included with Windows 11 Pro licensing and integrates with centralized management (Microsoft 365 Defender portal) at low additional cost — an efficient choice for a startup that still wants centralized visibility across all 20 endpoints.

Password Policy

A minimum 12-character password length, mandatory use of upper/lowercase letters, numbers, and symbols, no password reuse across the last 5 passwords, mandatory multi-factor authentication (MFA) for email and remote access, and account lockout after 5 failed login attempts. MFA in particular addresses the most common cause of account compromise — credential theft — at very low cost to implement.

Expansion Plan

As ABC Startup Solutions grows beyond 20 employees, the infrastructure should scale in phases: (1) add a second switch and additional access points to cover new office space, (2) migrate select workloads (e-mail, backups, or a staging environment) to the cloud to avoid large upfront hardware purchases, (3) introduce a dedicated backup appliance once data volume outgrows the NAS, and (4) formalize IT roles by hiring a dedicated Network Administrator once the current IT team of 5 can no longer cover both development and infrastructure duties. Planning these phases now prevents costly, reactive re-architecture later.

Part 8 — Personal Reflection

Working through this infrastructure plan for ABC Startup Solutions taught me that system administration begins long before any cable is plugged in or any server is racked. The biggest lesson was that every technical decision — how many desktops to buy, which switch to use, how to lay out the network — has to trace back to a business reason. It is not enough to know what a firewall does; I had to justify why this company, at this size, with this budget, needed one configured a specific way. That shift from "what tool works" to "what does the business actually need" was the most valuable part of the exercise.

The most challenging task was the network diagram. It is one thing to list network equipment in a table, and another to actually trace how traffic logically moves from the internet, through the firewall, into the switch, and out to four different departments without gaps or contradictions in the design. I found myself redrawing the flow more than once after realizing a department was connected in a way that bypassed the firewall entirely — a mistake that would be a real security hole in an actual company. Building the diagram forced me to think like an attacker as well as an administrator, checking whether my own design had holes in it.

This project also made clear why planning must come before deployment. If ABC Startup Solutions had simply purchased computers and signed up for internet service without this exercise, they likely would have ended up with mismatched hardware, no security policy, and no path to scale. Planning surfaces problems on paper, where they cost nothing to fix, instead of in production, where they cost money, downtime, and sometimes company data. Writing out the backup strategy alone made me realize how many small businesses operate with no real recovery plan until it is too late.

Overall, this project pushed me to practice the full workflow of a Junior System Administrator: gathering requirements, designing infrastructure, documenting decisions clearly enough for a non-technical executive to approve, and defending each choice with a reason rather than a guess. I feel more confident that I can walk into an entry-level IT role and not just execute tasks, but understand why those tasks matter to the business. Going forward, I want to get hands-on practice actually configuring the equipment I designed here — setting up a real firewall rule set, testing a real backup restore — so that this planning skill is backed by execution skill as well.